



Faculdade de Design,
Tecnologia e Comunicação
 Universidade Europeia

Project 1 - CICLA
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Master in Creative Computing &
Artificial Intelligence (IADE 2025–26)

MILESTONE 2

Prototype Development

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EXECUTIVE SUMMARY

Following Milestone 1, the project evolved from a single product concept into a modular B2B smart parking system. This shift was driven by collaboration with two master teams and the need to support different design outcomes through a shared technological core.

Milestone 2 focuses on validating this system-level approach through a working IoT prototype, data-driven intelligence, and interactive visualizations.





ABOUT CICLA

CICLA is a Portuguese project that works to make bike ownership safer by registering and identifying bicycles, helping to reduce theft and create trust in urban cycling. With Eco Pod, CICLA expands into physical smart city solutions for public space, starting in Lisbon with 15 pods.

ABOUT PROJECT CONTEXT & SCOPE

This project is part of a **multi-disciplinary collaboration** between three Master's programs at IADE:

- **Creative Computing & Artificial Intelligence**, responsible for IoT tech, data visualisation, AI, and digital interaction.
- **Product, Space and Interaction Design**, physical structure, materials, space, environmental integration.
- **Design and Advertisement**, visual identity, communication & branding towards target audience and the local government as the stakeholder.



ABOUT MILESTONE 2

This milestone (M2) focuses on **phase 2, prototyping and technical exploration**. As the creative computing & AI team, we translate the Eco Pod concept into a functional system by integrating the MVP, data flows, and core technical components. This milestone documents how AI logic, machine learning experiments, IoT sensing, and interactive interfaces come together to validate feasibility and form the foundation for the initial prototype.



SYSTEM CONCEPT & STRATEGY

Core Concept

CORE CONCEPT — FROM ECOPOD TO BICLANET

EcoPod was born from a simple observation: cyclists in Lisbon don't just need a place to park, they need a reason to trust it.

Designed primarily for daily commuters, EcoPod transforms ordinary street parking into a connected experience where technology builds confidence in safety and reliability.

By combining IoT sensors, live feedback, and contextual data, EcoPod reassures users that their bike is safely locked and that space is available before they arrive.

In this way, EcoPod acts as a digital layer of trust in everyday parking, a small but meaningful step toward safer, smarter urban cycling.

As the project progressed and collaboration expanded to multiple master teams, the team identified the need to **abstract this concept beyond a single physical product**. This led to the development of BiclaNet.

BiclaNet is a modular, white-label smart bicycle parking system designed to support multiple physical pod designs through a shared digital core.

Rather than focusing on one fixed product, BiclaNet prioritizes **data flow, system intelligence, and interaction reliability**, allowing different partners to build upon the same technological foundation.

At its core, BiclaNet preserves EcoPod's original intent: building trust.

Through real-time sensing, predictive logic, and clear digital feedback, the system helps cyclists feel confident that parking is available, their bike is secure, and conditions are suitable before arrival. Through a configurator website and dashboard the client is now expanded towards creating a back office system that can cater to multiple businesses.

This system-first approach allows the Creative Computing & AI team to focus on their core competencies; IoT integration, AI logic, machine learning, and interface design while enabling multiple physical, branding, and urban design outcomes to coexist within the same platform.

SYSTEM CONCEPT & STRATEGY

System Overview

System Overview

The BiciNet system is designed around a small set of core data flows that connect physical infrastructure, digital interfaces, and intelligent processing. Rather than attempting to model every possible interaction, this milestone focused on identifying and implementing the minimum set of data flows required to establish system reliability and confidence.

Core Data Flows (Milestone 2)

For the MVP, the system centers on three primary data flows:

1. Pod Occupancy
 - a. Indicates whether a parking slot is available, occupied, or in transition. This data originates from physical sensing at the pod level and is surfaced across both operator and rider interfaces.
2. Lock Status
 - a. Represents the real-time state of the locking mechanism (open, closed, opening, closing). This flow is critical for user reassurance and operational monitoring, linking hardware behavior directly to digital feedback.
3. Weather Context (via API)
 - a. External weather data is incorporated to provide contextual awareness within the system. This information supports AI-driven insights related to comfort, reliability, and predicted usage patterns.

Rationale for Scope Selection

These data flows were intentionally selected because they:

- Directly support both rider confidence and operator decision-making
- Are feasible to prototype and validate within the scope of Milestone 2
- Create a clear bridge between physical sensing, digital interfaces, and AI/ML layers

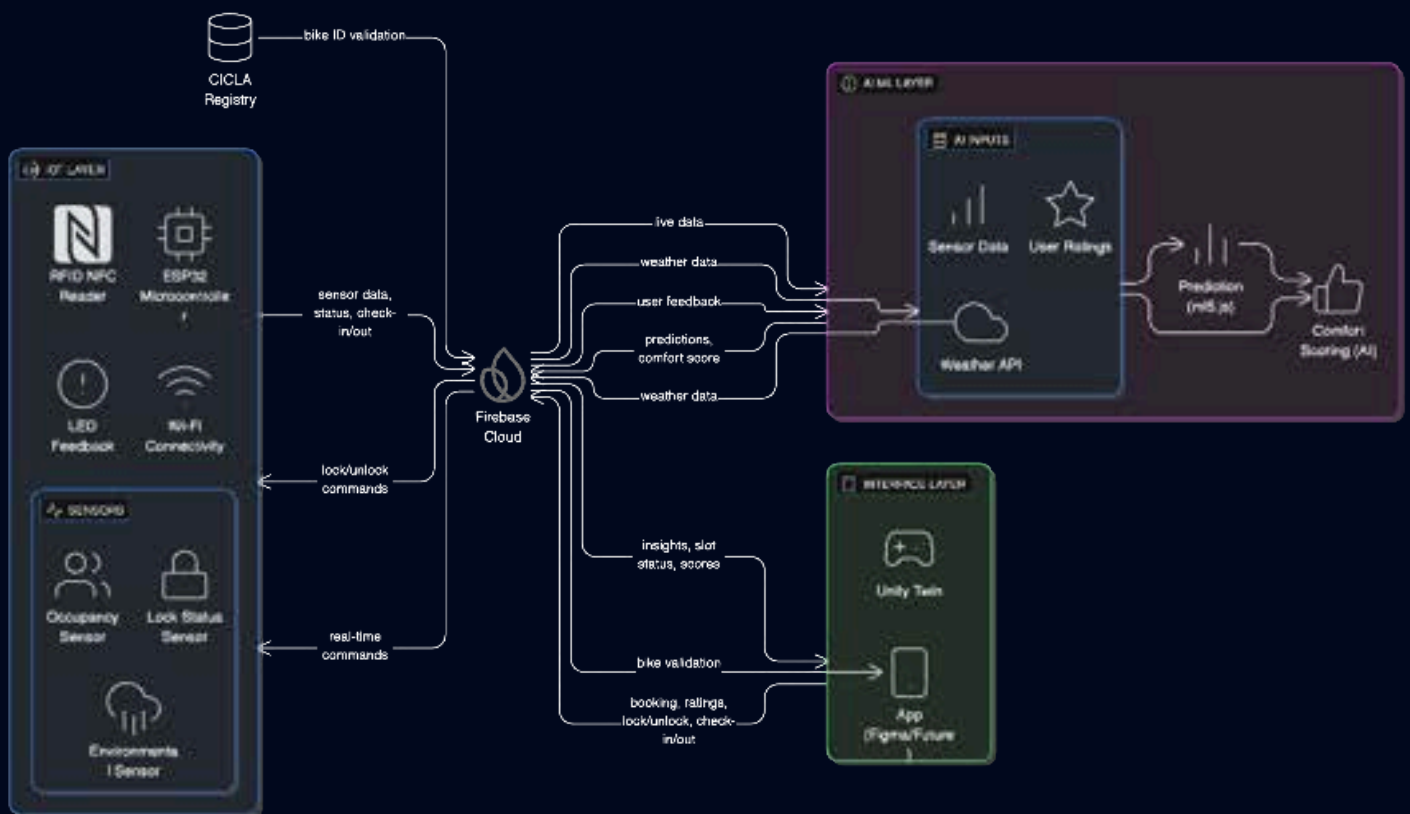
By anchoring the system around these flows, the project establishes a stable foundation upon which additional features and intelligence can be layered in future iterations. This system-level focus ensures that design, hardware, and software components remain aligned as the platform evolves.

SYSTEM CONCEPT & STRATEGY

System-Level MVP Definition

Eco Pod is structured across three complementary layers:

IoT, AI/ML, and Interface



Instead of implementing full barometric and wind sensing, Eco Pod integrates weather APIs.

This reduces hardware cost and complexity while still allowing us to use weather conditions as an input to our AI models.

Additional sensors, such as light or temperature, can be integrated later as the physical design and prototyping evolve and budget allows.

SYSTEM CONCEPT & STRATEGY

Core Data Flows

WHAT ABOUT DATA?

The initial data flow establishes how EcoPod connects physical sensing, environmental context, and user reassurance into one smart system. Occupancy, lock status, and weather data form the foundation for intelligent interaction, detecting activity, predicting availability, and adapting to conditions.

A fourth data flow in the future, user feedback, will complete the loop by adding community trust data to refine comfort scoring and improve reliability.

Data Flow 1: occupancy level of pods

Shows live pod usage and feeds the Machine Learning predictor with pod history (historical occupancy data, time of day, real time weather conditions affecting pod).

(#Physical Computing, ML, Creative Programming)

IoT sensor (load cell/ IR) → ESP32 → Firebase → App + ML model

Data Flow 2: lock status of parking slots

Provides visible reassurance to users (LED + app icon).

(#Physical Computing, AI, Creative Programming)

RFID/NFC input → Lock sensor (reed/limit switch) → ESP32 → Firebase



**App UI
& AI logic**



LED RGB output

SYSTEM CONCEPT & STRATEGY

Core Data Flows

Data Flow 3: weather data

Adds environmental awareness to predictability & comfort.
(# Introduction to AI, Applied ML)

Weather API (or simulated data) → AI Comfort Score / ML Feature

Data Flow 4 : user data (optional)

Integrates community trust and emotional safety into system intelligence;
not MVP, initial core concept but planned for later.
(#AI, Creative Programming)

App user input / pod rating system → Firebase → AI Comfort Score

User data history → Firebase → AI with ML Comfort Score

CONCEPT DEVELOPMENT & PROTOTYPING

Creative Programming

USER APP- DESIGN EXPLORATION (FRONT OFFICE)



The rider-facing application was first developed as a comprehensive set of interface designs in Figma to explore the full end-to-end user experience. These designs focused on mapping core flows, edge cases, and system states before committing to implementation.

The design work includes screens for:

- Onboarding and account setup
- Sign-in and authentication
- Main navigation and map-based discovery
- Pod selection and reservation
- Lock interaction and confirmation
- Failure and error states

Rather than optimizing for visual polish alone, the goal of this design phase was to clarify system behavior, define how real-world infrastructure states would be communicated to users, and ensure that moments of uncertainty (such as availability, errors, or failed interactions) were explicitly addressed.

Given the number of screens and variations required to fully represent these flows, the complete design system and screen set are maintained in a separate Figma file, shared alongside the report. This allowed the report to remain legible while still documenting the full scope of the application design.

Link to the Figma Designs:

<https://www.figma.com/design/SDHCGPVD4NOB0F975JNMXXK/BICLANET-MILESTONE2-MOCKUPS?NODE-ID=0-1&T=JRIQV761GNRGDREB-1>

CONCEPT DEVELOPMENT & PROTOTYPING

Creative Programming

USER APP - CODED PROTOTYPE & AI/ML INTEGRATION



Following the design exploration, a coded prototype of the rider application was developed to test how the system would behave with live and simulated data. This prototype focuses on translating the design concepts into an interactive experience connected to the BiclNet system architecture.

The coded prototype demonstrates:

- Map-based discovery of nearby pods
- Real-time availability and occupancy indicators
- AI/ML-driven insights surfaced directly in the interface
- Dynamic messaging related to system confidence and comfort

Unlike the static design mockups, this prototype integrates simulated IoT data and machine learning outputs to explore how predictive information could be presented to riders. For example, availability forecasts and comfort indicators are derived from usage patterns rather than fixed values, illustrating how uncertainty can be reduced before a rider arrives at a pod.

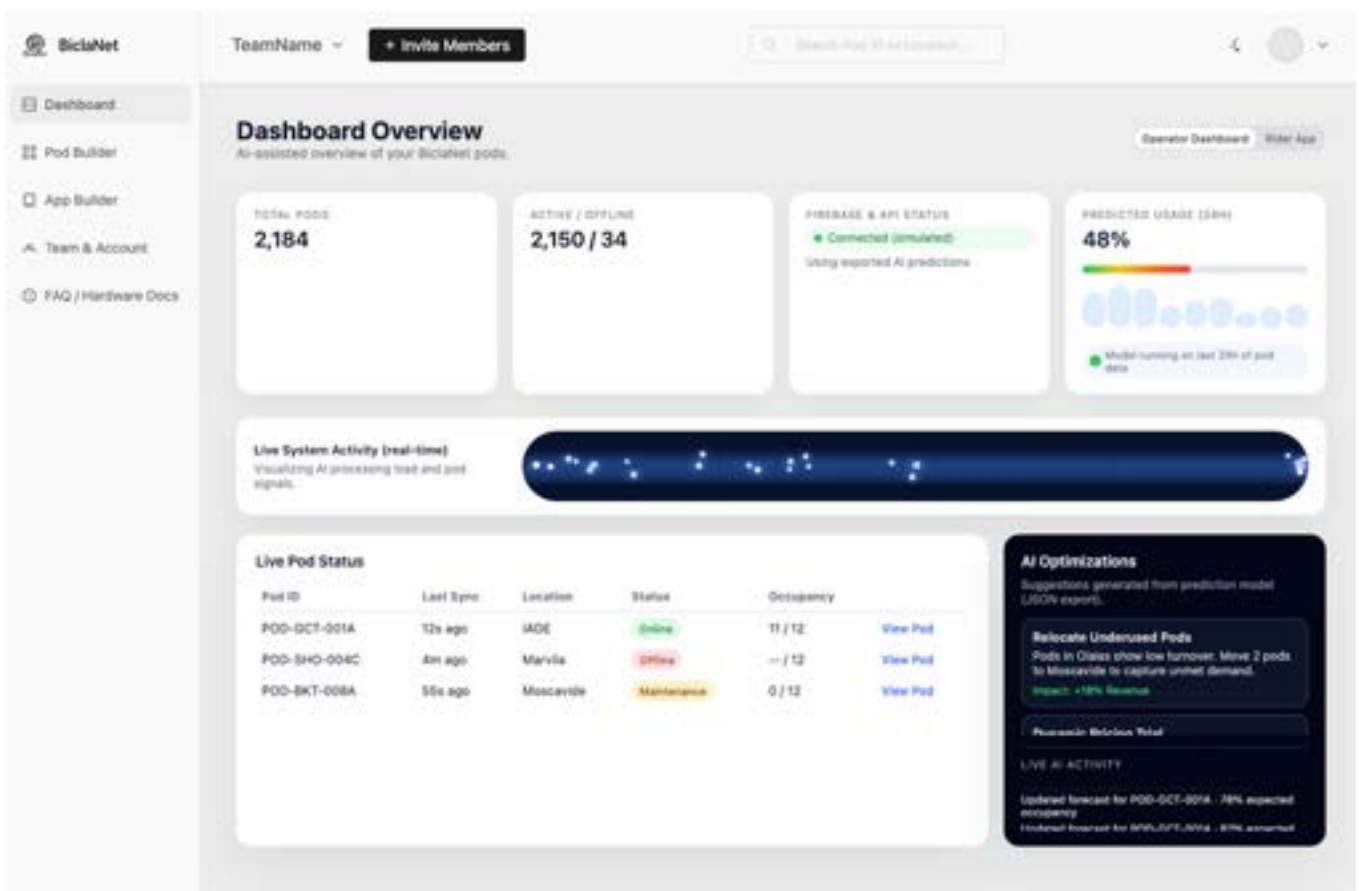
This separation between design mockups and a functional prototype allowed us to evaluate not only how the app looks, but how it behaves as part of a larger system, reinforcing the role of the rider app as an interface to infrastructure rather than a standalone product.

[HTTPS://NAYDINO.GITHUB.IO/BICLANET-AI-DEMO/APP.HTML](https://naydino.github.io/biclanet-ai-demo/app.html)

CONCEPT DEVELOPMENT & PROTOTYPING

Creative Programming

CLIENT DASHBOARD CONCEPT (BACK OFFICE)



[HTTPS://NAYDINO.GITHUB.IO/BICLANET-AI-DEMO/DASHBOARD.HTML](https://naydino.github.io/biclanet-ai-demo/dashboard.html)

The client dashboard prototype was developed to explore how BiclNet operators would monitor and manage infrastructure at scale. The interface combines real-time pod data, system health metrics, and AI-assisted insights into a single back-office view. This prototype demonstrates how physical infrastructure, cloud processing, and decision support tools are unified within the platform, and serves as the foundation for later refinements informed by user feedback.

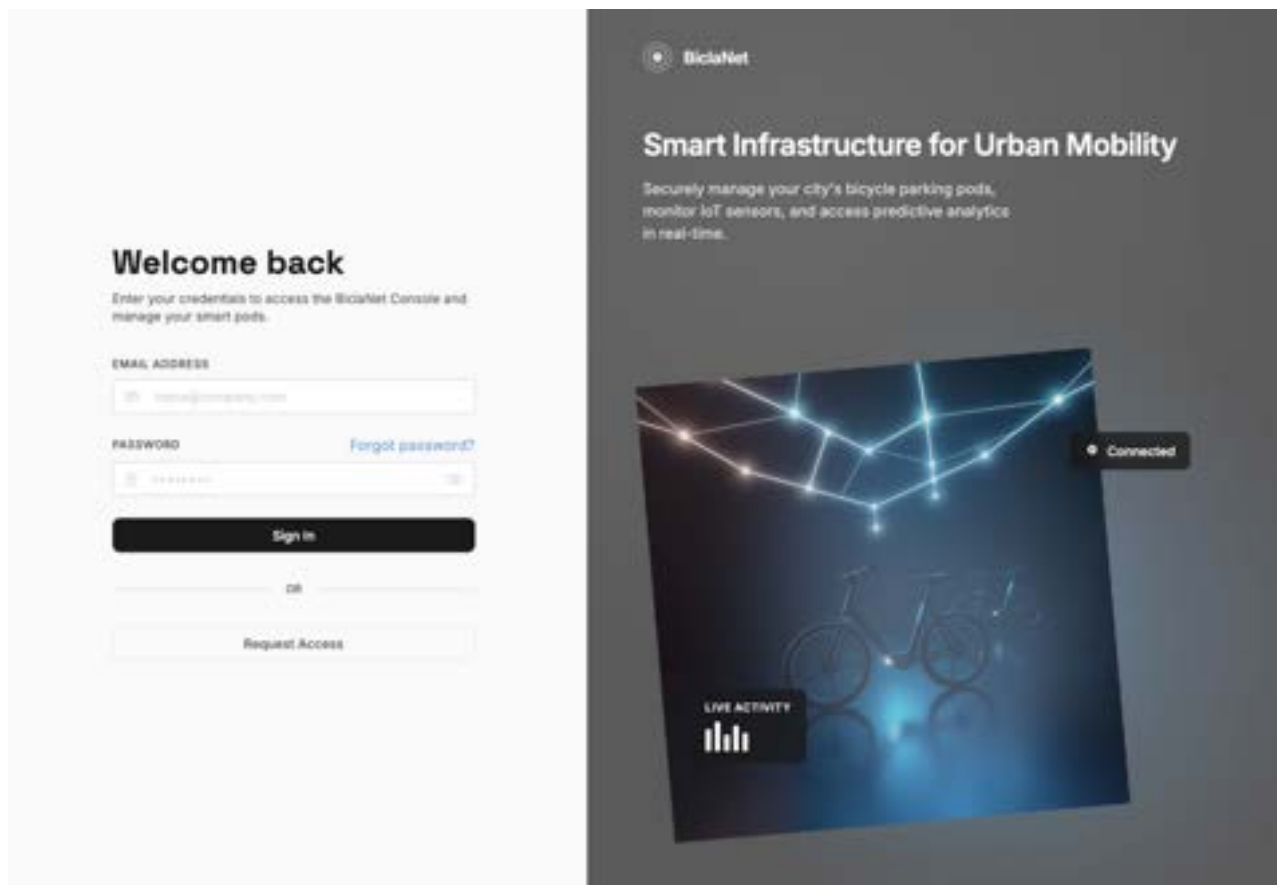
CONCEPT DEVELOPMENT & PROTOTYPING

User Testing Improvements

CLIENT WEBSITE

As part of Milestone 2, we developed a coded prototype of the BiciNet client-facing website to support collaboration with both the product design and marketing teams. While the initial focus of the project was on system architecture, app flows, and infrastructure configuration, interactions with other teams revealed a need for clearer communication and documentation at the platform level.

[HTTPS://NAYDINO.GITHUB.IO/BICLANET-AI-DEMO/](https://naydino.github.io/biclanet-ai-demo/)



RESPONDING TO CROSS-TEAM FEEDBACK

During discussions with the two product teams, it became clear that there was confusion around how the BiciNet system related to the physical pod designs. Team members frequently asked questions about required hardware, system responsibilities, and how the technology would integrate into their designs.

Similarly, conversations with the marketing teams highlighted a need for clearer high-level explanations of the platform's purpose and value. Rather than treating these questions as isolated misunderstandings, we recognized them as a signal that the system itself needed better outward-facing documentation.

CONCEPT DEVELOPMENT & PROTOTYPING

User Testing Improvements

FAQ & HARDWARE DOCUMENTATION PAGE



[HTTPS://NAYDINO.GITHUB.IO/BICLANET-AI-DEMO/FAQ.HTML](https://naydino.github.io/biclanet-ai-demo/faq.html)

In response, we created a dedicated FAQ & Hardware Documentation page within the client website. This page serves as a central reference for operators and partners, outlining:

- The role of BiclaNet within the overall infrastructure system
- Required and optional hardware components
- How sensors, locking mechanisms, and connectivity modules fit into different pod designs
- The division of responsibility between physical infrastructure and digital systems

This documentation helped align understanding across teams and reinforced the idea of BiclaNet as a modular, white-label platform rather than a fixed physical product.

CLIENT HOMEPAGE FOR PROSPECTIVE OPERATORS



We also introduced a client homepage to support prospective operators and partners. This page provides a concise overview of how the system works, the benefits of the platform, and the relationship between the physical pods, cloud system, and applications. It functions as an entry point for new clients, helping set expectations before deeper technical engagement.

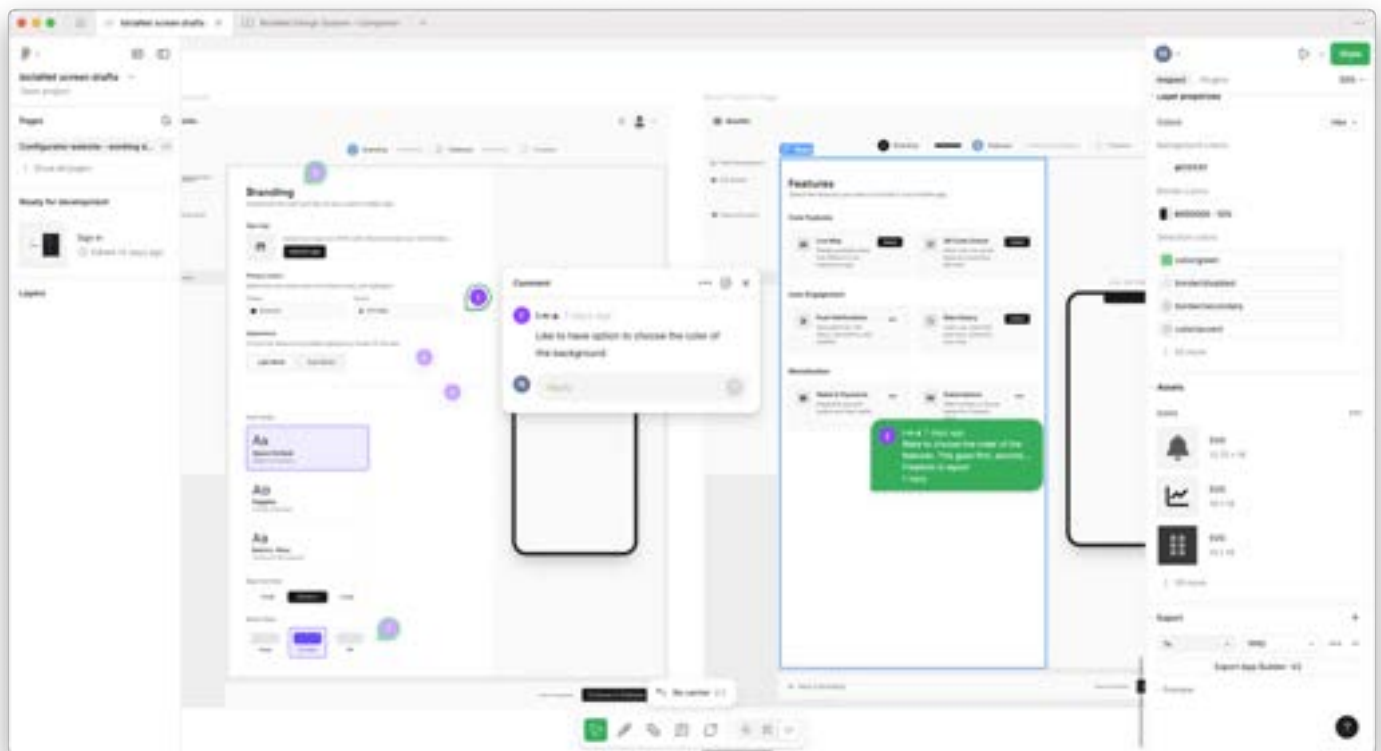
[HTTPS://NAYDINO.GITHUB.IO/BICLANET-AI-DEMO/](https://naydino.github.io/biclanet-ai-demo/)

CONCEPT DEVELOPMENT & PROTOTYPING

User Testing Improvements

APP BUILDER, REFINEMENT THROUGH USER FEEDBACK

As the App Builder interface evolved, we conducted informal user testing sessions with members of the marketing teams to better understand how non-technical stakeholders would interact with the configuration process.



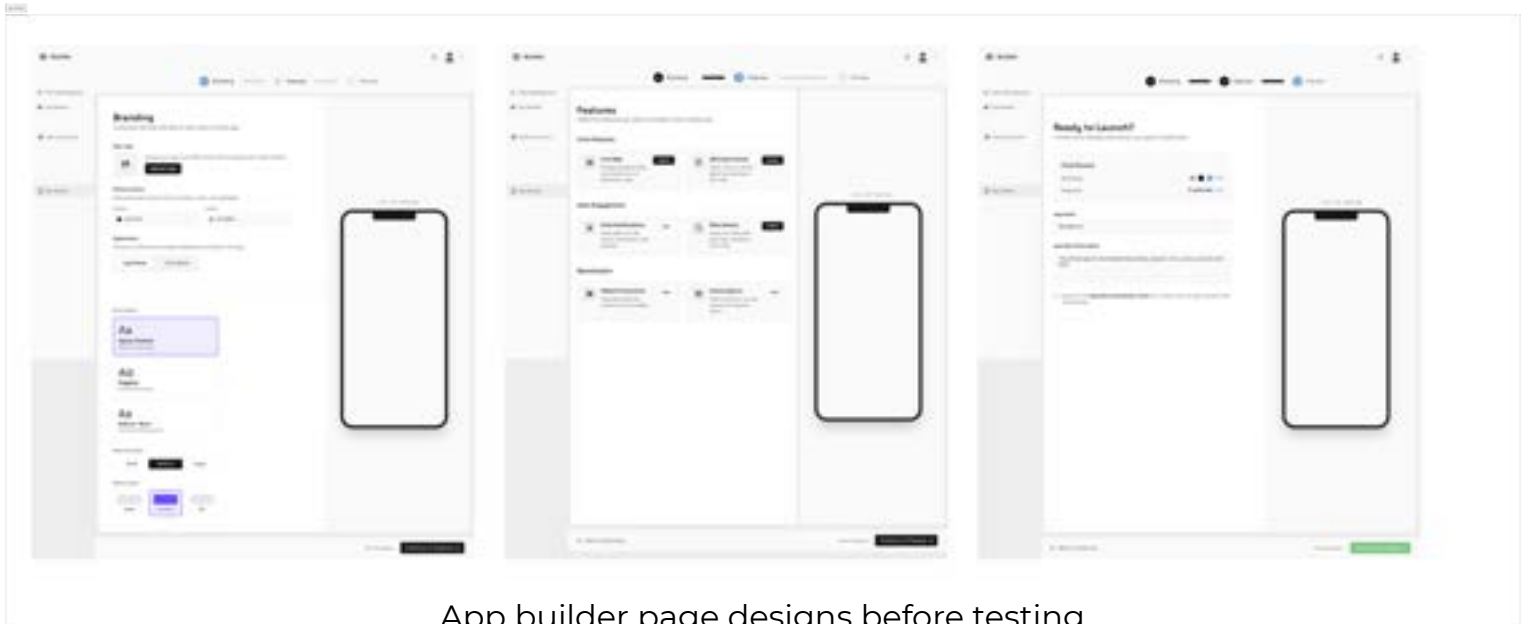
*screenshot of the user testing notes in Figma

CONCEPT DEVELOPMENT & PROTOTYPING

User Testing Improvements

APP BUILDER, REFINEMENT THROUGH USER FEEDBACK

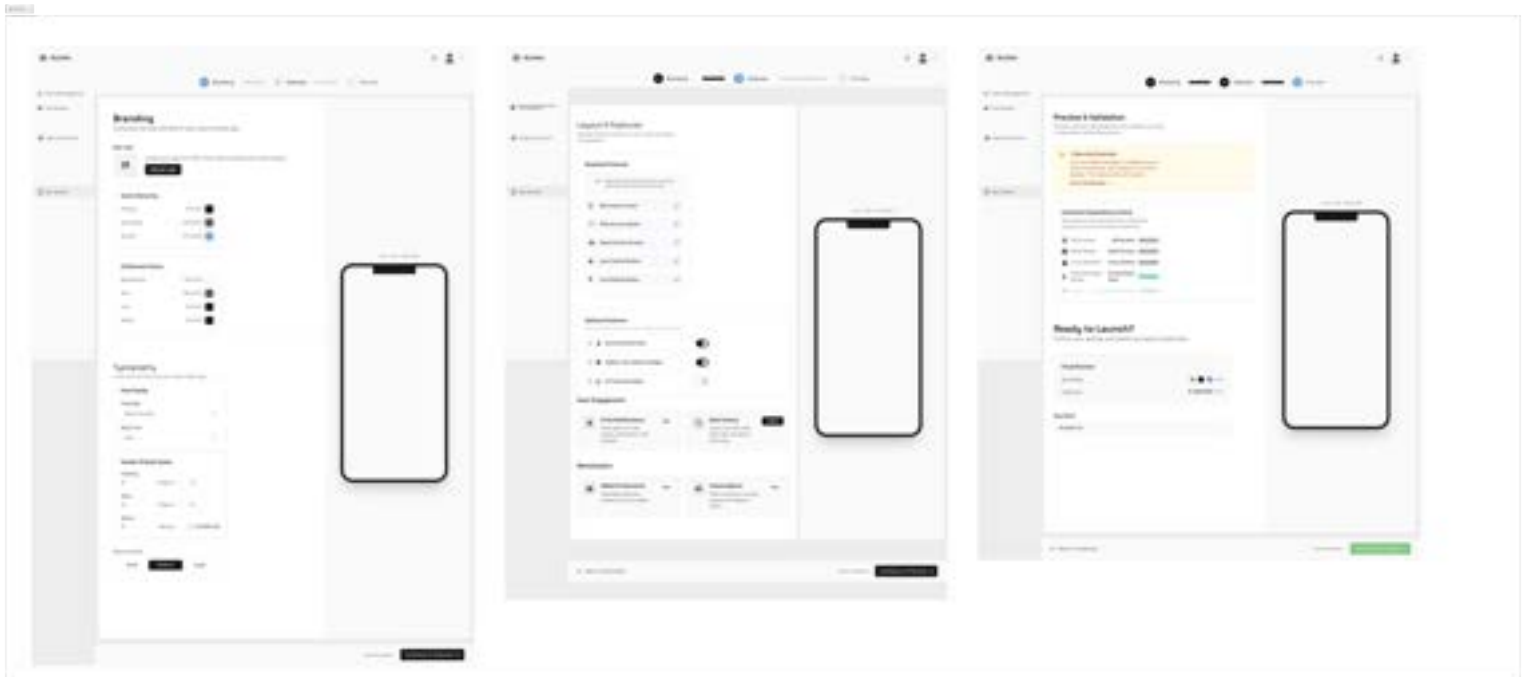
Based on this feedback, we refined the interface to allow more granular control over color hierarchy and feature ordering, while maintaining system constraints tied to hardware and functionality. These refinements improved flexibility for branding teams without compromising the consistency of the underlying system.



App builder page designs before testing



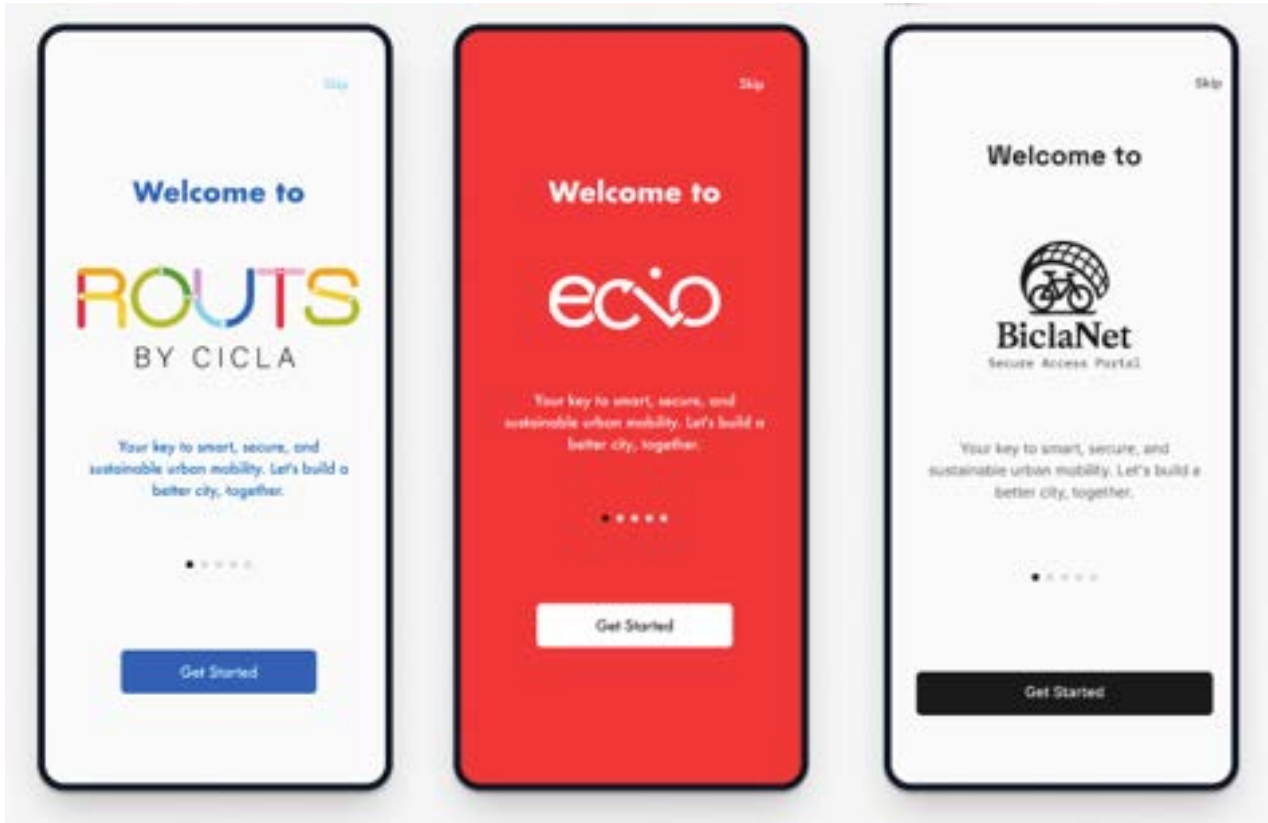
App builder page designs after testing



CONCEPT DEVELOPMENT & PROTOTYPING

White-Label Branding & Identity Adaptation

WHITE-LABEL APP CUSTOMIZATION



To support BiclaNet's B2B white-label strategy, we designed the rider app as a single adaptable system rather than a collection of one-off applications. Using the App Builder, the same core app structure can be customized to reflect different client and marketing identities through changes to color palettes, typography, and branding assets.

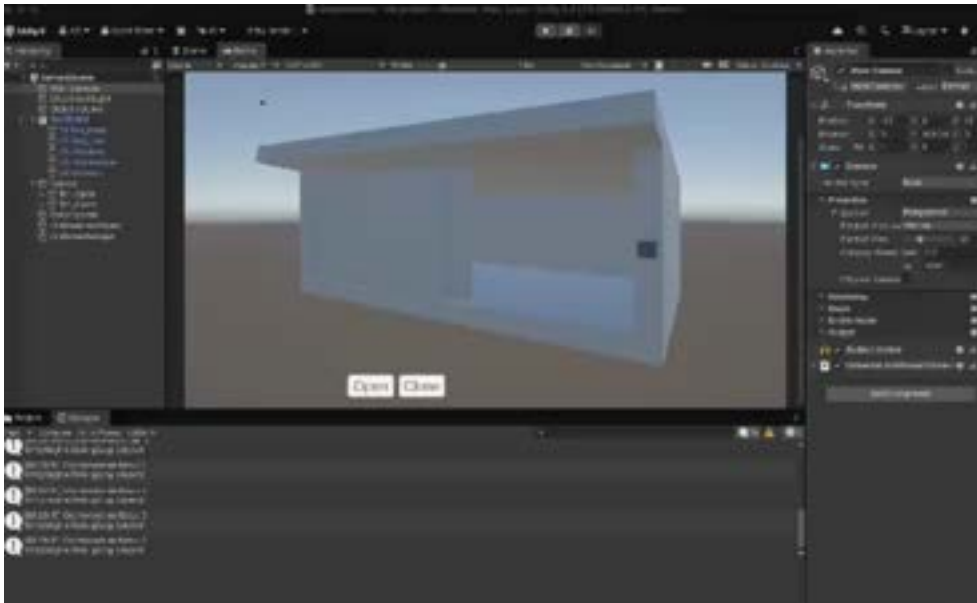
The examples shown demonstrate how a single underlying app experience can be visually adapted to align with multiple partner brands while preserving consistent functionality and interaction patterns. This approach allows marketing teams to maintain their own visual language and brand voice without requiring changes to the system logic or development of separate applications.

By separating brand identity from system behavior, the platform supports flexibility at the surface level while ensuring reliability and consistency underneath. This reinforces BiclaNet's role as infrastructure rather than a fixed consumer product, enabling adoption across different organizations, cities, and design languages.

One system, multiple brand identities. without duplicating apps or logic.

PROTOTYPES & EXPERIMENTS

Unity Digital Twin



Unity-Based Pod Simulation

In parallel with the physical IoT prototype, a Unity-based simulation was developed to visualize pod behavior and test interaction logic in a controlled environment. The simulation represents a digital twin of the physical pod, allowing operators to trigger open and close actions through a simple interface.

This simulation supports:

- Visual validation of pod interaction logic
- UI testing for operator controls
- A fallback demonstration environment when physical hardware is unavailable

While simplified, the Unity simulation complements the hardware prototype by enabling rapid testing of system states and reinforcing the system-level architecture of BiclaNet.

PROTOTYPES & EXPERIMENTS

IoT Prototype

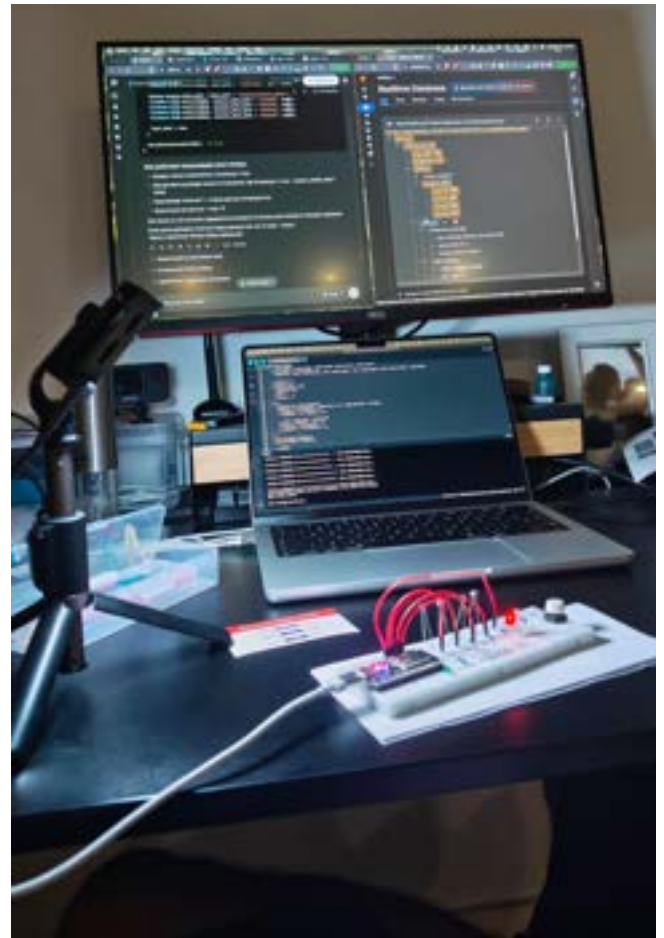
Physical IoT Prototype (ESP32 + Firebase)

A functional IoT prototype was developed to validate the physical-to-digital data flow of the BiciNet system. The prototype uses ESP32 microcontrollers to simulate pod hardware behavior and communicate real-time status updates to a shared Firebase database.

The prototype demonstrates:

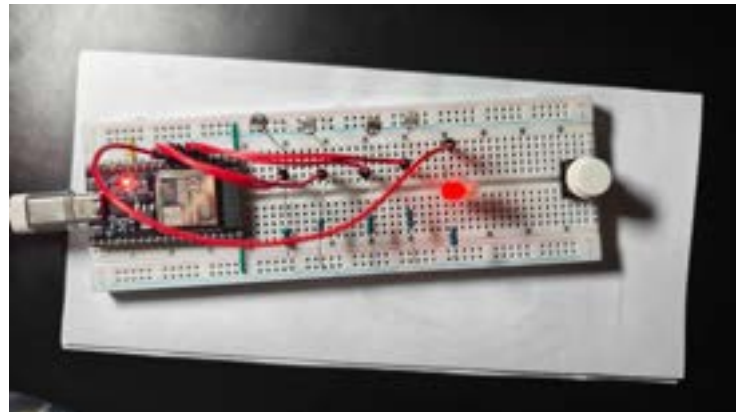
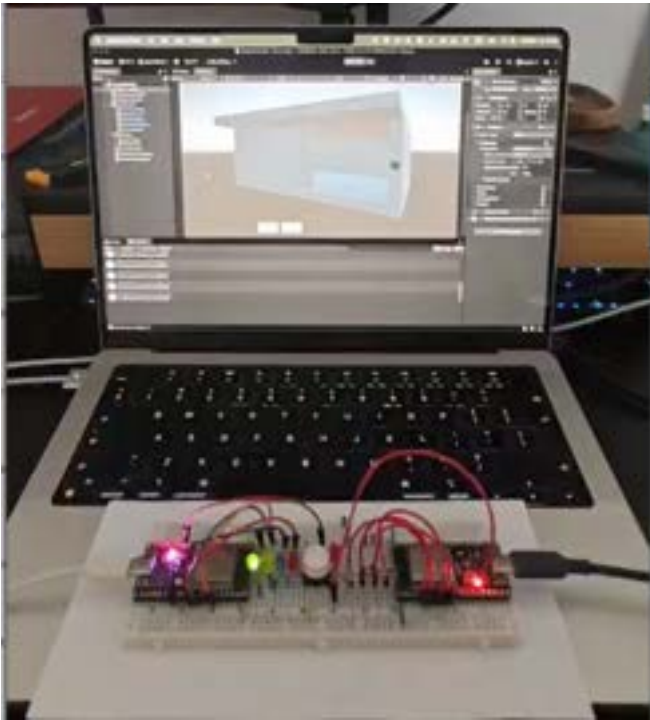
- Pod lock state changes (open / closed / opening / closing)
- Occupancy sensing using simulated sensor inputs
- Visual status feedback through multi-color LEDs
- Bidirectional communication between hardware, cloud database, and applications

Rather than representing a final mechanical solution, this prototype acts as a data-generating system designed to test sensing, communication reliability, and system integration. This approach allowed rapid iteration while remaining aligned with future physical pod designs developed by the product teams.



PROTOTYPES & EXPERIMENTS

IoT Prototype



The IoT prototype and hardware integration.

The work focused on:

- Implementing ESP32 microcontrollers connected to a Firebase Realtime Database
- Synchronizing multiple ESP32 devices to represent pod-level and slot-level behavior
- Implementing push-buttons and LED indicators to represent lock state transitions
- Sending simulated sensor data (e.g., temperature / slot activity) to the system

This work validated the feasibility of real-time communication between physical infrastructure and the BiclNet platform, forming the foundation for future hardware refinement and scaling.

REFLECTION & NEXT STEPS

Deferred Ideas & Design Decisions

Advanced Interaction & Accessibility Concepts

Several features were intentionally deferred to future milestones, including camera-based theft detection, biometric access, and advanced environmental sensing. These ideas were explored conceptually but postponed due to ethical considerations, technical complexity, and scope constraints. Deferring these elements allowed the team to focus on validating core system feasibility during Milestone 2 and adapting to a new core concept.

During early ideation, we explored more advanced interaction models for smart bicycle parking infrastructure that go beyond standard app-based or button-based controls. One such concept was the development of a “claw” holder prototype, a mechanically adaptive capture system designed to accommodate different bicycle types, frame geometries, and wheel sizes through multi-axis adjustment ML-class Proposal

This proposal also investigated contactless interaction methods, including:

- Hand gesture recognition using camera-based hand pose detection
- Voice commands for hands-free operation
- Hybrid control options combining mobile app input, physical controls, and manual adjustment

These ideas were motivated by real user challenges identified early in the project, such as difficulty stabilizing a bicycle while interacting with controls, accessibility needs, and concerns about phone battery dependency during use ML-class Proposal

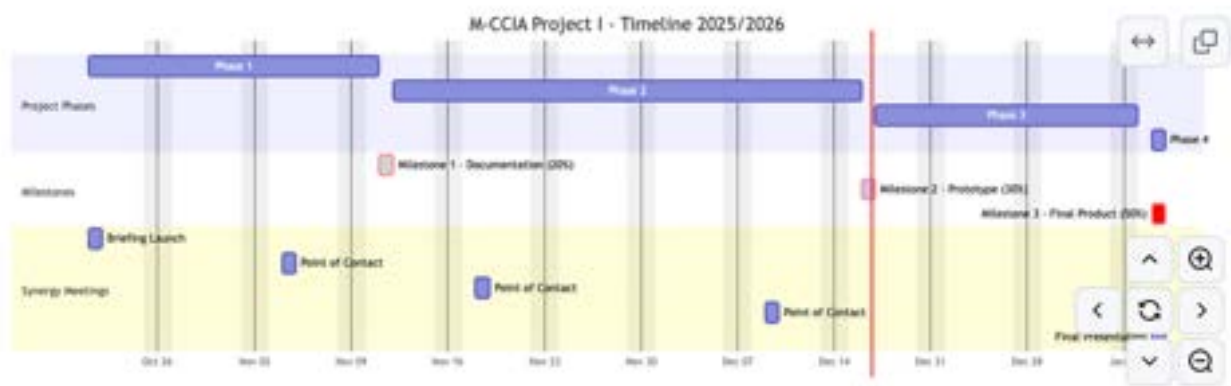
From a technical perspective, the proposal outlined the potential use of ml5.js models (hand pose detection and speech classification) integrated with a Firebase-backed system to enable gesture- and voice-driven interaction ML-class Proposal.

Ultimately, these concepts were intentionally deferred for Milestone 2. While promising, they introduced significant mechanical complexity, ethical considerations, and implementation overhead that would have diverted focus from validating the core BiclaNet system architecture. Instead, the team prioritized establishing a reliable foundation for real-time data flow, configurability, and platform-level integration, ensuring that such advanced interaction models could be explored meaningfully in future iterations.

Activity Plan & Execution Record

ROADMAP AND TASKS

Project Timeline – Gantt Chart



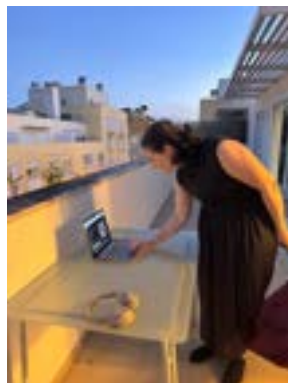
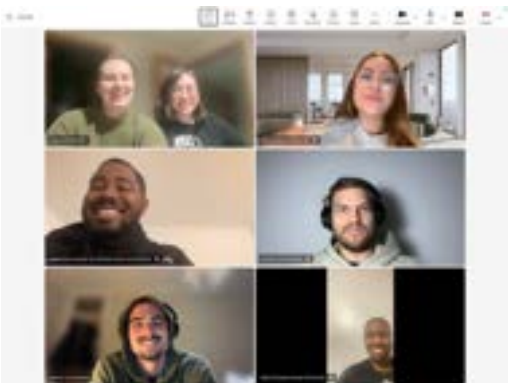
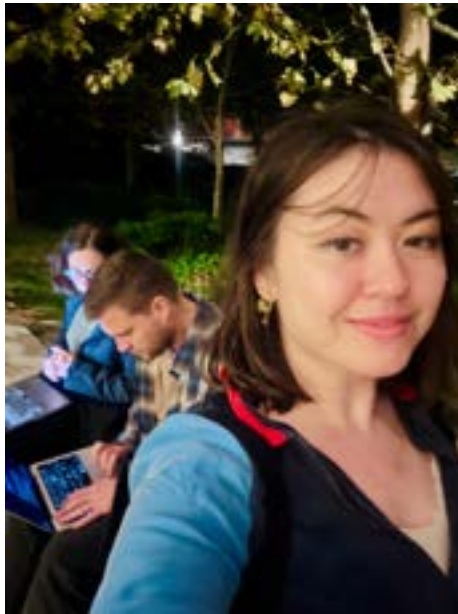
The project followed a phased structure aligned with the course milestones. While an initial planning framework was defined, the execution evolved iteratively in response to concept changes, team restructuring, and collaboration with additional master groups. Key activities were tracked through team meetings, point of contacts with the other teams, shared documents, and GitHub repositories.

Phase	Period	Main Activities	Notes
Phase 1	Oct–Nov	Research, concept definition, Milestone 1	Initial EcoPod concept
Phase 2	Nov–Dec	Prototype exploration, system abstraction	Pivot to BiclaNet
Phase 3	Dec–Jan	MVP prototyping with system integration, documentation	Milestone 2 delivery

During the phase 2 and leading up to milestone 2, execution was impacted by unforeseen constraints including team restructuring and periods of limited availability, requiring prioritization of core deliverables.

TEAM

Teamwork & Synergy



LIST OF REQUIREMENTS

Functional Requirements

Use Case	Step-by-Step Flow (HOW the user interacts)
1. Register & Log In	1. Open app → "Continue with Google" or "Apple ID" (OAuth) 2. App receives ID token → verifies with backend via HTTPS 3. First-time user: Consent screen ("Allow location for locker discovery?") → Save preferences locally 4. Home screen loads with map of nearby lockers
2. Find & Reserve a Locker	1. Map auto-centers via coarse location (Wi-Fi + cell) 2. Tap locker icon → shows real-time slots (via BLE beacon broadcast) 3. Tap "Reserve 30 min" → app sends encrypted reservation token to locker via BLE 4. Locker LED blinks green + push: "Slot 3 reserved!"
3. Load Bike (Lock)	1. Arrive at locker → open app → "Open Slot 3" 2. App activates camera → scan QR on slot door 3. BLE handshake: app ↔ locker (challenge-response) 4. Solenoid unlocks → vibration + sound feedback 5. User places bike → closes door → weight sensor confirms 6. Locker auto-locks → app shows "Secured until 18:00"
4. Unload Bike (Unlock)	1. Return to locker → app detects geofence (50 m) → shows "Unlock" button 2. Tap "Unlock" → app sends one-time unlock code (expires in 60 s) via BLE 3. Door opens → proximity sensor detects removal 4. App confirms: "Bike released. Session ended."
5. Rent a Bike (from shared pool)	1. Tap "Rent Bike" → see available e-bikes on map 2. Select bike → scan NFC tag on handlebar 3. App unlocks motor lock via BLE 4. Ride starts → app tracks via GPS + accelerometer (low-power mode) 5. End ride: park in any locker → auto-lock → payment deducted
6. Offline Mode	1. No internet → app caches last 10 reservations (encrypted SQLite) 2. BLE still works → user can unlock reserved slot 3. On reconnect: delta sync (only changed records)

LIST OF REQUIREMENTS

Non-Functional Requirements

1. Privacy & GDPR Compliance	
Requirement	How it's Solved
Minimal data collection	Only collect: email (from OAuth), coarse location, reservation timestamps. No precise GPS track unless renting.
User consent	First launch: granular toggles ("Share location for map?" / "Allow analytics?"). Can revoke anytime.
Right to be forgotten	"Delete Account" → app wipes local data + sends DELETE /user to backend → confirmed in < 5 s.
Data encryption	All BLE payloads: AES-256-GCM with per-session keys. Local storage: SQLCipher.
No server-side tracking	Backend logs anonymized hashes only (no IP, no device ID).
2. Low Energy Consumption	
Strategy	Implementation
BLE Low Energy only	Use advertising mode (100 ms interval) when idle. Connection only on user action.
Smart location	Coarse location (Wi-Fi) → geofence trigger (50 m radius) → wake app → switch to fine GPS only if needed.
Sensor duty cycle	Weight & door sensors: sleep 99% of time, wake on BLE command.
Result	< 2% battery drain per 8-hour day
3. Sustainability	
Aspect	Solution
Hardware	Lockers: recycled aluminum, modular battery pack (CR123A, user-replaceable).
E-bikes	Swappable LG 21700 cells, repairable frames.
End-of-life	QR code → "Recycle here" links to local e-waste centers.

LIST OF MATERIALS

Component	Purpose / Function	Qty.	Estimated Unit Price (€)	Est. Subtotal (€)
ESP32 Dev Board	Main IoT controller with Wi-Fi/BLE to send data to Firebase / app	1	10 – 15	10 – 15
Load cell + HX711 or IR distance sensor	Detects whether a parking slot is occupied	2	7 – 10	14 – 20
Limit switch / magnetic reed sensor	Confirms when the lock is properly closed	2	2	4
RC522 module (or similar)	Reads bike-centric tag for secure access / check-in out	1	8 – 12	8 – 12
Cards / stickers / keyfobs	Linked to the user's bike ID (CICLA registry)	4 – 6	1.5 – 2	6 – 12
Push button switch	Manual check-in/out for prototype demo	1 – 2	1	1 – 2
RGB LED strip (12 V) or high-brightness LEDs	Indicates slot status: free / occupied / locked	1 strip	10 – 12	10 – 12
Mini servo motor + simple latch or 3D printed bar	Demonstrates smart-lock opening / closing	1	8 – 12	8 – 12
Breadboard + jumper wires + resistors kit	Connects and tests components safely	1 set	6 – 8	6 – 8
12 V adapter + step-down converter to 5 V	Powers RGB LED strip and ESP32 circuit	1 each	6 – 8	6 – 8
DC jack, Dupont wires, USB cable	General connectivity and power	1 set	3 – 5	3 – 5

→ **Estimated total (for 2 bike slots): €76 – 110**

ADDITIONAL LINKS

[Figjam WorkFile](#)

[App Mockup](#)

[Figma Mockups](#)

[Github Repository](#)

[Client Website Prototype](#)

[Github Project Plan](#)